

WHICH REAL QUADRATIC PLANES NEED FOUR COLOURS

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ABSTRACT. For a subfield $F \subset \mathbb{R}$ let $\chi(F^2)$ be the chromatic number of the graph on F^2 that joins two points at Euclidean distance 1. For a real quadratic field $\mathbb{Q}(\sqrt{d})$, d squarefree, it is known that $\chi(\mathbb{Q}(\sqrt{d})^2) = 2$ unless $d \equiv 3 \pmod{4}$, that $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 3$ if $d \equiv 3 \pmod{4}$, and that $\chi(\mathbb{Q}(\sqrt{d})^2) \leq 3$ unless $d \equiv 11 \pmod{12}$. We prove that $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every $d \equiv 11 \pmod{12}$. So the plane over $\mathbb{Q}(\sqrt{d})$ needs four colours exactly when $d \equiv 11 \pmod{12}$, and, with the upper bounds of Fischer and Moorhouse, $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ for all these d except possibly $d \equiv 47, 143, 167 \pmod{168}$. The proof exhibits, for each d , a finite set of unit vectors, made of $(1, 0)$, a vector and its mirror image, one more vector when $d \equiv 11 \pmod{24}$, and their rotations by the rational rotations whose denominators divide 5^k for a suitable k , and shows that no character of the group they generate maps all of them into $[1/3, 2/3]$; a characterisation of 3-colourable Cayley graphs of abelian groups by such characters then excludes 3-colourings. The main step is an exact description of the characters of $\frac{1}{N}\mathbb{Z}[i]$, $N = 5^k$, that keep every rational rotation with denominator dividing N in $[1/3, 2/3]$.

1. INTRODUCTION

Let $F \subset \mathbb{R}$ be a field. The *plane over F* is the graph on F^2 in which x and y are adjacent when $|x - y| = 1$, and $\chi(F^2)$ is its chromatic number. For $F = \mathbb{R}$ this is the Hadwiger–Nelson problem, and $6 \leq \chi(\mathbb{R}^2) \leq 7$: the bound $\chi(\mathbb{R}^2) \geq 5$ is due to de Grey [3] and to Exoo and Ismaïlescu [4], the bound $\chi(\mathbb{R}^2) \geq 6$ to a recent non-constructive proof [10], and the upper bound to Isbell (see [14]). For $F = \mathbb{Q}$, $\chi(\mathbb{Q}^2) = 2$ [16]. For real quadratic fields $K = \mathbb{Q}(\sqrt{d})$, $d \geq 2$ squarefree, the following was known:

- $\chi(K^2) = 2$ if $d \not\equiv 3 \pmod{4}$ [6, 5, 8];
- $\chi(K^2) \geq 3$ if $d \equiv 3 \pmod{4}$ [5, Thm. 8] (and [8, Thm. 8.6] when K contains $\sqrt{p}$ for a prime $p \equiv 3 \pmod{4}$); for $u = ((1 - d)/(1 + d), 2\sqrt{d}/(1 + d))$ and its mirror image $\bar{u}$, the relation $\frac{d+1}{4}(u + \bar{u}) + \frac{d-1}{2}(1, 0) = 0$ is a closed walk of odd length d ;
- $\chi(K^2) \leq 3$ if $d \not\equiv 2 \pmod{3}$ [5, Thm. 9], [8, Cor. 8.3];
- $\chi(K^2) \leq 4$ if $d \equiv 3 \pmod{8}$ [5, Thm. 10] or $d \equiv 0, 1, 2, 4 \pmod{7}$ [8, Cor. 8.3]; Moorhouse’s Theorem 8.1 combines his bounds at 3 and 7 with his Lemma 8.4: $\chi(K^2) \leq 4$ unless $d \equiv 47, 59, 83 \pmod{84}$.

So a real quadratic field can need four colours only when $d \equiv 11 \pmod{12}$. In the literature we found, the largest lower bound for a real quadratic field is 3 [7, 8, 5]; explicit graphs for 27 fields that need four colours are given in the companion draft [12]. Moorhouse asked what can be said about $\chi(\mathbb{Q}(\sqrt{d})^2)$, and about $\chi(\mathbb{Q}(\sqrt{47})^2)$ in particular [9]. Fischer [5, Thm. 10(ii)] showed, with the vector u above (up to sign and mirror image), that for $d \equiv -1 \pmod{24}$ the vectors $(\frac{1}{2}, 0)$ and $(\frac{1}{3}, 0)$ are sums of unit vectors, so that the plane has no *additive* k -colouring for $k \leq 6$: none induced by a homomorphism from the group C_0 generated by the unit vectors to a group of order k (his Theorem 1 extends colourings of C_0 to the plane by cosets).

Theorem 1. $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ for every positive integer $d \equiv 11 \pmod{12}$.

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Corollary 2. *For squarefree $d \geq 2$, $\chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$ if and only if $d \equiv 11 \pmod{12}$. For these d , $\chi(\mathbb{Q}(\sqrt{d})^2) = 4$ unless $d \equiv 47, 143, 167 \pmod{168}$; in particular $4 \leq \chi(\mathbb{Q}(\sqrt{47})^2) \leq 5$.*

Proof. The first statement is Theorem 1 and the bounds listed above. Among $d \equiv 11 \pmod{12}$, those not covered by Fischer's or Moorhouse's bound 4 satisfy $d \equiv 7 \pmod{8}$ and $d \equiv 3, 5, 6 \pmod{7}$, that is, $d \equiv 47, 143, 167 \pmod{168}$. For $d = 47$, the bound 5 comes from reduction at a prime above 11, which splits in $\mathbb{Q}(\sqrt{47})$, and $\chi(\mathbb{F}_{11}^2) = 5$ [8, Lemma 8.2 and Table 6.1]. $\square$

Corollary 3. *For every prime $p \geq 5$, $\chi(\mathbb{Q}_p^2) \geq 4$, where $\mathbb{Q}_p^2$ is the graph joining $x, y \in \mathbb{Q}_p^2$ when $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 1$. So $\chi(\mathbb{Q}_p^2) \geq 4$ exactly for $p \geq 5$.*

Proof. As $\gcd(p, 12) = 1$, some positive integer d satisfies $d \equiv 11 \pmod{12}$ and $d \equiv 1 \pmod{p}$ (it need not be squarefree; Theorem 1 covers it). By Hensel's lemma d has a square root in $\mathbb{Z}_p^\times$, so $\sqrt{d} \mapsto s$ defines a field embedding $\sigma: \mathbb{Q}(\sqrt{d}) \rightarrow \mathbb{Q}_p$. Being an injective ring homomorphism fixing $\mathbb{Q}$, σ preserves the relation $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 1$ in both directions, so applied coordinatewise it maps the plane over $\mathbb{Q}(\sqrt{d})$ isomorphically onto an induced subgraph of $\mathbb{Q}_p^2$, and $\chi(\mathbb{Q}_p^2) \geq \chi(\mathbb{Q}(\sqrt{d})^2) \geq 4$. For $p = 2, 3$ no $d \equiv 11 \pmod{12}$ is a square in $\mathbb{Q}_p$, and indeed $\chi(\mathbb{Q}_2^2) = 2$ and $\chi(\mathbb{Q}_3^2) = 3$, following from [7]. $\square$

For $p \equiv 1 \pmod{4}$ this is weaker than $\chi(\mathbb{Q}_p^2) = \infty$, a consequence of a theorem of Davies [1]; the new case is $p \equiv 3 \pmod{4}$, where explicit graphs over 27 fields had given the bound for every such p with $7 \leq p < 2\,129\,503\,819$ but not for $p = 2\,129\,503\,819$, and where no finite set of fields can give it for every p , by Chebotarev's theorem [11]. A finite 4-chromatic unit-distance graph in $\mathbb{Q}_p^2$ exists, by compactness, but is not exhibited.

The proof is explicit. Let G_N be the set of the $4(2k+1)$ rational unit vectors whose denominators divide $N = 5^k$ (Lemma 5), and for an integer n let

$$u_n = \frac{(n + i\sqrt{d})^2}{n^2 + d} = \left(\frac{n^2 - d}{n^2 + d}, \frac{2n\sqrt{d}}{n^2 + d} \right),$$

a unit vector of $\mathbb{Q}(\sqrt{d})^2$, with mirror image $\bar{u}_n$. Rotating a vector v by $\gamma \in G_N$ gives a unit vector γv of $\mathbb{Q}(\sqrt{d})^2$.

Theorem 4. *Let $d \equiv 11 \pmod{12}$, $N = 5^k$, and let U be the following set of unit vectors.*

- (1) *If $d \equiv 23 \pmod{24}$ and $d < 21 \cdot 5^{k-1}$ (for example $5^k \geq d/4$): $U = G_N \cup G_N u_1 \cup G_N \bar{u}_1$.*
- (2) *If $d \equiv 11 \pmod{24}$, $s = v_3(d+1)$, n is a positive integer with $n \equiv 3 \pmod{4}$ and $n \equiv 1 \pmod{3^{s+1}}$ (for example $n = 1 + 2 \cdot 3^{s+1}$), and $5^k > \frac{2\sqrt{10}}{3} n(n+d)$: $U = G_N \cup G_N u_1 \cup G_N \bar{u}_1 \cup G_N u_n$.*

Then no homomorphism ξ from the group $\mathbb{Z}U$ generated by U to $\mathbb{R}/\mathbb{Z}$ maps every vector of U into the arc $[1/3, 2/3]$.

Proof of Theorem 1. By [15, Theorem 1], a Cayley graph $\text{Cay}(\Gamma, S)$ of an abelian group Γ , with $S = -S$, is 3-colourable if and only if some character $\xi: \Gamma \rightarrow \mathbb{R}/\mathbb{Z}$ maps S into $[1/3, 2/3]$. By Theorem 4, $\text{Cay}(\mathbb{Z}U, \pm U)$ is not 3-colourable. It is a subgraph of the plane over $\mathbb{Q}(\sqrt{d})$. (If d is not squarefree, its squarefree part is again $\equiv 11 \pmod{12}$, and the field is the same.) $\square$

By the theorem of de Bruijn and Erdős [2], each of these planes contains a finite unit-distance graph with no proper 3-colouring, and deleting vertices one at a time gives one with chromatic number exactly 4; it is triangle-free, since $\sqrt{3} \notin \mathbb{Q}(\sqrt{d})$. We do not exhibit it: the direction “3-colourable $\Rightarrow$ character” of [15, Theorem 1] averages over the group with an invariant mean and is not constructive. For 27 of these fields, explicit vertex-critical graphs with 71 to 1404 vertices and no proper 3-colouring are given in [12], with proofs that do not use characters.

Section 2 turns characters into complex numbers, Section 3 describes the set S_N of characters of $\frac{1}{N}\mathbb{Z}[i]$ that are allowed on G_N , and Sections 4 and 5 prove the two cases of Theorem 4. The bound on k is generous: for $d = 11$ the theorem needs $5^k = 3125$, but $5^k = 25$ (40 vectors up to sign) already works, as an exact certificate checked by computer shows (Section 6).

2. CHARACTERS AS COMPLEX NUMBERS

Identify $\mathbb{Q}^2$ with $\mathbb{Q}(i)$, and write a vector of $\mathbb{Q}(\sqrt{d})^2$ as $X + \sqrt{d}Y$ with $X, Y \in \mathbb{Q}(i)$: the vector $(X_1 + \sqrt{d}Y_1, X_2 + \sqrt{d}Y_2)$ corresponds to $X = X_1 + iX_2, Y = Y_1 + iY_2$. For $\gamma \in \mathbb{Q}(i)$ with $|\gamma| = 1$, multiplying X and Y by γ is the rotation by γ , an isometry of $\mathbb{Q}(\sqrt{d})^2$. In this notation $u_n = X + \sqrt{d}Y$ with $X = (n^2 - d)/(n^2 + d)$ and $Y = 2ni/(n^2 + d)$.

Let U be one of the sets of Theorem 4 and T the least common multiple of the denominators $n^2 + d$ of the vectors u_n used. Every vector of U lies in $\Gamma' = M + \sqrt{d}M$, $M = \frac{1}{NT}\mathbb{Z}[i]$, because $N\gamma \in \mathbb{Z}[i]$ for $\gamma \in G_N$ and $u_n = X + \sqrt{d}Y$ with $X, Y \in \frac{1}{T}\mathbb{Z}[i]$. Every character ξ of $\mathbb{Z}U$ extends to Γ' , since $\mathbb{R}/\mathbb{Z}$ is divisible. The characters of $\Gamma' \cong \mathbb{Z}^4$ are $X + \sqrt{d}Y \mapsto \text{Re}(\bar{c}_1 X) + \text{Re}(\bar{c}_2 Y) \pmod{1}$ with $c_1, c_2 \in \mathbb{C}$ (on the basis $\frac{1}{NT}, \frac{i}{NT}$ of M these take the arbitrary real values $\frac{\text{Re } c}{NT}, \frac{\text{Im } c}{NT}$). For a vector $v = X + \sqrt{d}Y$ put $w_v = c_1 \bar{X} + c_2 \bar{Y} \in \mathbb{C}$. Then

$$(1) \quad \xi(\gamma v) = \text{Re}(\bar{w}_v \gamma) \pmod{1} \quad (\gamma \in G_N),$$

and $v \mapsto w_v$ is $\mathbb{Q}$ -linear, so every linear relation with rational coefficients among vectors holds, exactly in $\mathbb{C}$, among the corresponding numbers w_v . Let $I = [1/3, 2/3]$ and

$$S_N = \{c \in \mathbb{C} : \text{Re}(\bar{c}\gamma) \in I + \mathbb{Z} \text{ for every } \gamma \in G_N\}.$$

If $\xi(U) \subseteq I + \mathbb{Z}$, then $w_v \in S_N$ for each $v \in \{1, u_1, \bar{u}_1, u_n\}$, where $1 = (1, 0)$. The set S_N is invariant under $N\mathbb{Z}[i]$ (as $N\gamma \in \mathbb{Z}[i]$), under $c \mapsto -c$ and under multiplication by i .

3. THE SHAPE OF S_N

The definitions of G_N and S_N make sense for every $k \geq 0$; $G_1 = \{\pm 1, \pm i\}$. Let $\rho = (3 + 4i)/5 = (2 + i)/(2 - i)$, $c_N^* = N(1 + i)/2$, $Q_N = \{\frac{N}{3}(\alpha + \beta i) : \alpha, \beta \in \{1, 2\}\}$, $\text{Sq}_0 = \{x + iy : |x|, |y| \leq 1/6\}$, and for $k \geq 0$

$$P_k = \{x \in \mathbb{C} : \bar{x}\rho^j \in \text{Sq}_0 \text{ for all integers } |j| \leq k\},$$

a closed convex polygon; $P_0 = \text{Sq}_0$ and $P_k \subseteq P_1$ for $k \geq 1$. Let $\Lambda_{\pm} = \frac{2 \pm i}{5}\mathbb{Z}[i]$; both contain $\mathbb{Z}[i]$ with index 5.

Lemma 5. $G_N = \{\varepsilon \rho^j : \varepsilon \in \{\pm 1, \pm i\}, |j| \leq k\}$, a set of $4(2k + 1)$ elements.

Proof. Let $\gamma \in G_N$, $\gamma = \alpha/5^t$ with $\alpha \in \mathbb{Z}[i]$ and $t \leq k$ least. Then $|\alpha|^2 = 25^t$, so $\alpha = \varepsilon(2 + i)^e(2 - i)^f$ with $e + f = 2t$; if $t \geq 1$, minimality excludes $e, f \geq 1$, so $\gamma = \varepsilon(2 \pm i)^{2t}/5^t = \varepsilon \rho^{\pm t}$. These elements are distinct, since ρ^j has exact denominator $5^{|j|}$. $\square$

Lemma 6. (a) $\bar{c}_N^* \gamma \in \frac{1+i}{2} + \mathbb{Z}[i]$ for every $\gamma \in G_N$. Hence $c_N^* + x \in S_N$ if and only if $\bar{x}\gamma \in \text{Sq}_0 + \mathbb{Z}[i]$ for every $\gamma \in G_N$; in particular $c_N^* + P_k \subseteq S_N$. (b) $Q_N \subseteq S_N$.

Proof. (a) By Lemma 5, $\bar{c}_N^* \gamma = \frac{5^{k-|j|}}{2}(1 - i)\varepsilon\pi$ with $\pi = (2 \pm i)^{2|j|}$. As $\varepsilon\pi$ has odd norm, $\varepsilon\pi \equiv 1 \pmod{1 + i}$ and $(1 - i)\varepsilon\pi \equiv 1 - i \pmod{2}$, so both coordinates are odd. Since $I = \frac{1}{2} + [-\frac{1}{6}, \frac{1}{6}]$, the conditions for γ and $i\gamma$ together say $\bar{x}\gamma \in \text{Sq}_0 + \mathbb{Z}[i]$. (b) For $q = \frac{N}{3}(\alpha + \beta i)$, $\bar{q}\gamma = \frac{5^{k-|j|}}{3}(\alpha - \beta i)\varepsilon\pi$. In $\mathbb{Z}[i]/3 = \mathbb{F}_9$, the elements with both coordinates nonzero, $\pm 1 \pm i$, are the non-squares, since $1 + i$ generates $\mathbb{F}_9^\times$ and $(1 + i)\mu_4 = \{\pm 1 \pm i\}$. Now $\alpha - \beta i$ is a non-square and $\varepsilon\pi$ a nonzero square, so both coordinates of $(\alpha - \beta i)\varepsilon\pi$ are prime to 3, and those of $\bar{q}\gamma$ lie in $\{\frac{1}{3}, \frac{2}{3}\} + \mathbb{Z}$. $\square$

Lemma 7. (a) $\max_{x \in P_1} |x| = \sqrt{10}/18 \approx 0.1757$. (b) Every point of $(\Lambda_+ \cup \Lambda_-) \setminus \mathbb{Z}[i]$ is at distance at least $\sqrt{2}/6 \approx 0.2357$ from $\text{Sq}_0 + \mathbb{Z}[i]$, and none lies in $[-\frac{1}{3}, \frac{1}{3}]^2 + \mathbb{Z}[i]$.

Proof. (a) P_1 is cut out by the six slabs $|\langle x, \nu \rangle| \leq \frac{1}{6}$, $\nu \in \{1, i, \rho, i\rho, \bar{\rho}, i\bar{\rho}\}$, whose directions modulo 180° are $0^\circ, \theta, 90^\circ - \theta, 90^\circ, 90^\circ + \theta, 180^\circ - \theta$ with $\theta = \arccos \frac{4}{5}$. So P_1 is a 12-gon circumscribed about the circle of radius $\frac{1}{6}$; consecutive normals make angles θ (four times) and $90^\circ - 2\theta$ (twice), and the vertices are at distance $\frac{1}{6}/\cos \frac{\theta}{2} = \frac{1}{6} \cdot \frac{\sqrt{10}}{3}$, for example $(\frac{1}{6}, \frac{1}{18})$, or closer. (b) The nonzero classes of $\Lambda_+/\mathbb{Z}[i]$ are those of $\pm \frac{2+i}{5}$ and $\pm \frac{-1+2i}{5}$, whose representatives in $(-\frac{1}{2}, \frac{1}{2})^2$ have one coordinate $\pm \frac{2}{5}$ and the other $\pm \frac{1}{5}$. The point of Sq_0 nearest to $\frac{2}{5} + \frac{i}{5}$ is $\frac{1}{6} + \frac{i}{6}$, at distance $\sqrt{(7/30)^2 + (1/30)^2} = \sqrt{2}/6$; other translates of Sq_0 are farther. The coordinate $\pm \frac{2}{5}$ is at distance $\frac{2}{5} > \frac{1}{3}$ from $\mathbb{Z}$. Λ_- is the mirror image of Λ_+ . $\square$

Lemma 8 (unique lifts). *Let $k \geq 1$, $N = 5^k$, $N' = 5^{k-1}$. If $c, c' \in S_N$ and $c' - c \in N'\mathbb{Z}[i]$, then $c' - c \in N\mathbb{Z}[i]$.*

Proof. Write $c' = c + N'\lambda$. Then $\bar{c}'\rho^k - \bar{c}\rho^k = N'\bar{\lambda}\rho^k = \bar{\lambda}(2+i)^{2k-1}/(2-i) \in \Lambda_+$. Both $\bar{c}\rho^k$ and $\bar{c}'\rho^k$ lie in $(I + \mathbb{Z}) + i(I + \mathbb{Z})$ (the conditions for ρ^k and $i\rho^k$), so the difference lies in $[-\frac{1}{3}, \frac{1}{3}]^2 + \mathbb{Z}[i]$, hence in $\mathbb{Z}[i]$ by Lemma 7(b): that is, $(2-i) \mid \bar{\lambda}$, or $(2+i) \mid \lambda$. With ρ^{-k} and Λ_- , $(2-i) \mid \lambda$. So $5 \mid \lambda$. $\square$

Proposition 9. *For every $k \geq 1$, $S_{5^k} = (c_N^* + P_k + N\mathbb{Z}[i]) \cup (Q_N + N\mathbb{Z}[i])$. So every $c \in S_N$ is $c = e + x$ with either $e \in c_N^* + N\mathbb{Z}[i]$ and $|x| \leq \sqrt{10}/18$, or $e \in Q_N + N\mathbb{Z}[i]$ and $x = 0$.*

Proof. “ $\supseteq$ ” is Lemma 6. For “ $\subseteq$ ” we induct on k ; write $N' = 5^{k-1}$ and note that $S_N \subseteq S_{N'}$ and $c_N^* - c_{N'}^* = 2N'(1+i) \in N'\mathbb{Z}[i]$.

$k = 1$. $S_1 = \frac{1+i}{2} + \text{Sq}_0 + \mathbb{Z}[i]$, so $c \in S_5$ is $c = c_5^* + x + \lambda$ with $x \in \text{Sq}_0$, $\lambda \in \mathbb{Z}[i]$. By Lemma 6(a), the conditions at ρ say $\bar{x}\rho + \bar{\lambda}\rho \in \text{Sq}_0 + \mathbb{Z}[i]$, where $\bar{\lambda}\rho \in \Lambda_+$. If $\bar{\lambda}\rho \notin \mathbb{Z}[i]$, Lemma 7(b) gives $|x| = |\bar{x}\rho| \geq \sqrt{2}/6$, and as $x \in \text{Sq}_0$, x is a corner $(\pm 1 \pm i)/6$; likewise with $\bar{\rho}$. If x is not a corner, then $(2+i) \mid \lambda$ and $(2-i) \mid \lambda$, so $c \equiv c_5^* + x \pmod{5\mathbb{Z}[i]}$, and $\bar{x}\rho^{\pm 1} \in \text{Sq}_0 + \mathbb{Z}[i]$ with $|x| < \frac{5}{6}$ gives $\bar{x}\rho^{\pm 1} \in \text{Sq}_0$: $x \in P_1$. If x is a corner, $c \equiv \frac{1+i}{2} + \frac{\pm 1 \pm i}{6} = \frac{\alpha + \beta i}{3} \pmod{\mathbb{Z}[i]}$ with $\alpha, \beta \in \{1, 2\}$; the point $q = \frac{5}{3}(\alpha' + \beta' i) \in Q_5$ with $\alpha' \equiv 2\alpha$, $\beta' \equiv 2\beta \pmod{3}$ lies in S_5 and $q \equiv c \pmod{\mathbb{Z}[i]}$, so $c \in q + 5\mathbb{Z}[i]$ by Lemma 8.

$k \geq 2$. Let $c \in S_N \subseteq S_{N'}$. If $c \equiv \frac{N'}{3}(\alpha + \beta i) \pmod{N'\mathbb{Z}[i]}$, then $q = \frac{N}{3}(\alpha' + \beta' i) \in Q_N$ with $\alpha' \equiv 2\alpha$, $\beta' \equiv 2\beta \pmod{3}$ satisfies $q \equiv c \pmod{N'\mathbb{Z}[i]}$ and $q \in S_N$, so $c \in q + N\mathbb{Z}[i]$ by Lemma 8. Otherwise $c = c_N^* + x + N'\lambda$ with $x \in P_{k-1} \subseteq P_1$ and $\lambda \in \mathbb{Z}[i]$, and the conditions at ρ^k say $\bar{x}\rho^k + \mu \in \text{Sq}_0 + \mathbb{Z}[i]$ with $\mu = N'\bar{\lambda}\rho^k \in \Lambda_+$. If $\mu \notin \mathbb{Z}[i]$, Lemma 7 gives $\sqrt{2}/6 \leq |x| \leq \sqrt{10}/18$, which is impossible. So $(2+i) \mid \lambda$, likewise $(2-i) \mid \lambda$, hence $c \equiv c_N^* + x \pmod{N\mathbb{Z}[i]}$, and $\bar{x}\rho^{\pm k} \in \text{Sq}_0$, so $x \in P_k$. $\square$

Types. By Proposition 9 write each $c \in S_N$ as $c = N\varepsilon + x$ with $|x| \leq r := \sqrt{10}/18$ and ε of type c , $\varepsilon \in \frac{1+i}{2} + \mathbb{Z}[i]$, or of type q , $\varepsilon \in \frac{\alpha + \beta i}{3} + \mathbb{Z}[i]$ with $\alpha, \beta \in \{1, 2\}$; in both cases $N\varepsilon \in \frac{N}{6}\mathbb{Z}[i]$. Let $\mathbb{Z}_{(p)}[i]$ be the elements of $\mathbb{Q}(i)$ whose denominators are prime to p , and v_3 the 3-adic valuation of $\mathbb{Q}(i)$ (3 is inert in $\mathbb{Z}[i]$). Then: ε has type q if and only if $v_3(\varepsilon) = -1$; type c if and only if $\varepsilon \notin \mathbb{Z}_{(2)}[i]$, and then $\varepsilon \equiv h := \frac{1+i}{2} \pmod{\mathbb{Z}_{(2)}[i]}$; $mh \equiv h$ for odd m , and $2h \in \mathbb{Z}[i]$.

Lemma 10 (exact relations). *Let $\sigma_1, \dots, \sigma_\ell \in S_N$, $\sigma_j = N\varepsilon_j + x_j$, and let $m_1, \dots, m_\ell \in \mathbb{Z}$ with $\sum m_j \sigma_j = 0$. If $N > 6r \sum |m_j|$, then $\sum m_j \varepsilon_j = 0$.*

Proof. $\sum m_j N\varepsilon_j = -\sum m_j x_j$; the left side lies in $\frac{N}{6}\mathbb{Z}[i]$, whose nonzero elements have modulus at least $\frac{N}{6}$, and the right side has modulus at most $r \sum |m_j| < \frac{N}{6}$. $\square$

4. THE CASE $d \equiv 23 \pmod{24}$

Let $a = \frac{d+1}{4}$ and $b = \frac{d-1}{2}$: here $a \equiv 0 \pmod{6}$, b is odd and $3 \nmid b$. The relation $a(u_1 + \bar{u}_1) + b \cdot 1 = 0$ gives, for $z = w_1$, $w = w_{u_1}$, $w' = w_{\bar{u}_1}$ in S_N ,

$$a(w + w') + bz = 0.$$

By Proposition 9 write $w = e_1 + x_1$, $w' = e_2 + x_2$, $z = e_3 + y$, with $e_j \in (c_N^* + N\mathbb{Z}[i]) \cup (Q_N + N\mathbb{Z}[i])$, $x_1, x_2, y \in P_k$, and the corresponding x_j or y equal to 0 when $e_j \in Q_N + N\mathbb{Z}[i]$. Then

$$a(e_1 + e_2) + be_3 = -(ax_1 + ax_2 + by).$$

As a is even and $3 \mid a$, $ac_N^* = \frac{a}{2}N(1+i)$ and $a\frac{N}{3}(\alpha + \beta i)$ lie in $N\mathbb{Z}[i]$; as b is odd and $3 \nmid b$, be_3 lies in $c_N^* + N\mathbb{Z}[i]$ or in $Q_N + N\mathbb{Z}[i]$ with e_3 . So the left side $u + iv$ lies in $c_N^* + N\mathbb{Z}[i]$, where $|u|, |v| \geq \frac{N}{2}$, or in $Q_N + N\mathbb{Z}[i]$ ($-Q_N \equiv Q_N$), where $|u|, |v| \geq \frac{N}{3}$. As P_k is convex and symmetric, the right side lies in $(2a+b)P_k = dP_k$ in the first case and in $2aP_k = \frac{d+1}{2}P_k$ in the second ($y = 0$). On $P_1 \supseteq P_k$ the conditions at ρ and $\bar{\rho}$ give $|3u \pm 4v| \leq \frac{5}{6}$, so $3|u| + 4|v| \leq \frac{5}{6}$. Hence the first case needs $\frac{7N}{2} \leq \frac{5d}{6}$, that is $d \geq \frac{21N}{5}$, and the second needs $\frac{7N}{3} \leq \frac{5(d+1)}{12}$, that is $d+1 \geq \frac{28N}{5}$. If $d < 21N/5$, neither holds. This proves Theorem 4(1); and $5^k \geq d/4$ gives $d \leq 4N < 21N/5$.

Remark. For $k \leq 5$ the bound is sharp: if $d \geq 21N/5$, then $c_N^*/d \in P_k$, since for $k \leq 5$ the conditions at $\rho^{\pm 1}$ are the binding ones, and this gives a character with all three types c. An exact computation (cells of S_N , Minkowski sums and the lattice points of $N\mathbb{Z}[i]$, with a second method as a cross-check and an explicit character in every feasible case) confirms, for every $d \equiv 23 \pmod{24}$ up to 1295 when $N = 5, 25, 125$ and up to 2999 when $N = 625$, that the set of Theorem 4(1) admits no such character exactly when $d < 21N/5$. For $k \geq 6$ the bound is not sharp: as $(1-i)\rho^6 = (1457 - 22049i)/15625$, the point c_N^*/d lies in P_6 only when $d \geq 66147N/15625$, and for $N = 5^6$ an exact decision (all choices of types and all lattice points) finds no character for the 22 values $65\,639 \leq d \leq 66\,143$ with $d \equiv 23 \pmod{24}$, although $d \geq 21N/5 = 65\,625$; the first value with a character is 66 167.

5. THE CASE $d \equiv 11 \pmod{24}$

Now a is odd, and the argument fails: $a + b$ is even and $3 \mid a$, so $a(c_N^* + q) + bc_N^* \in N\mathbb{Z}[i]$ for $q \in Q_N$, and since $\gcd(a, b) = 1$ suitable lifts $z = c_N^* + N\lambda_3$, $w = c_N^* + N\lambda_1$, $w' = q + N\lambda_2$ solve $a(w + w') + bz = 0$ exactly. So $G_N \cup G_N u_1 \cup G_N \bar{u}_1$ admits such a character for every k . The vector u_n excludes it. Comparing X and Y gives $(n^2 + d)u_n = (n-1)(n+d) \cdot 1 + n(1+d)u_1$, so with $y = w_{u_n}$,

$$a(w + w') + bz = 0, \quad (n^2 + d)y - (n-1)(n+d)z - n(1+d)w = 0.$$

As $n \geq 1$, $(n-1)(n+d)$ and $n(1+d)$ are nonnegative, and the sums of their absolute values are $2a + b = d$ and $2n(n+d) \geq d$. (For negative n the second sum is larger, and the hypothesis on N would have to change.) As $N > \frac{2\sqrt{10}}{3}n(n+d) = 12r n(n+d)$, Lemma 10 applies to both relations: writing $z = N\varepsilon_z + x_z$ and so on,

$$(2) \quad a(\varepsilon_w + \varepsilon_{w'}) = -b\varepsilon_z, \quad (n^2 + d)\varepsilon_y = (n-1)(n+d)\varepsilon_z + n(1+d)\varepsilon_w,$$

exactly in $\mathbb{Q}(i)$.

Step 1: ε_z has type c and $v_3(\varepsilon_z) \geq s-1$. We have $v_3(a) = s \geq 1$, $3 \nmid b$, and $v_3(\varepsilon_w + \varepsilon_{w'}) \geq -1$, so $v_3(\varepsilon_z) \geq s-1 \geq 0$.

Step 2 (at 2): ε_y has type c if and only if ε_w has type q. As $d \equiv 3 \pmod{8}$ and $n \equiv 3 \pmod{4}$, each of $n^2 + d$, $(n-1)(n+d)$ and $n(1+d)$ is 4 times an odd integer; write the second relation as $A'\varepsilon_y = B'\varepsilon_z + C'\varepsilon_w$ with A', B', C' odd. Modulo $\mathbb{Z}_{(2)}[i]$, $B'\varepsilon_z \equiv h$, and $C'\varepsilon_w \equiv h$ or 0 according as ε_w has type c or q. So $A'\varepsilon_y \in \mathbb{Z}_{(2)}[i]$ exactly when ε_w has type c.

Step 3 (at 3): ε_y has type q if and only if ε_w has type q. Let $e = v_3(n-1) \geq s+1$; $3 \nmid n(n+1)$. Then $n^2 + d = (n-1)(n+1) + (d+1)$ and $n+d = (n-1) + (d+1)$ have valuation s , so do $n(1+d)$, and $v_3((n-1)(n+d)) = e + s$. Dividing the second relation by 3^s gives $A''\varepsilon_y = B''\varepsilon_z + C''\varepsilon_w$ with $3 \nmid A''C''$ and $v_3(B''\varepsilon_z) \geq e + s - 1 \geq 2s \geq 2$. So $v_3(\varepsilon_y) = -1$ if and only if $v_3(\varepsilon_w) = -1$.

Steps 2 and 3 contradict each other whatever the type of ε_w . This proves Theorem 4(2).

Remark. The argument of Section 4 does not work with one vector in place of u_1 and u_n : if $v = (\mu + i\nu\sqrt{d})/\tau$ with $\mu^2 + d\nu^2 = \tau^2$ in lowest terms and $d \equiv 3 \pmod{8}$, then $4 \nmid \tau$ (if τ is even, μ, ν are odd and $\mu^2 + d\nu^2 \equiv 4 \pmod{8}$), while the argument of Section 4 needs the coefficient $\tau/\gcd(\tau, 2\mu)$ of the relation $\tau(v + \bar{v}) = 2\mu \cdot 1$ to be even. Correspondingly, $x^2 + y^2$ is anisotropic over $\mathbb{Q}_2(\sqrt{d})$ for $d \equiv 3 \pmod{8}$ and isotropic for $d \equiv 7 \pmod{8}$.

6. CHECKS

Every computation that the proofs use is checked in exact arithmetic by a program in [13]: Lemmas 5 and 6 for $k \leq 6$, both numbers of Lemma 7, and S_{5^k} itself for $k = 1, 2, 3$ by exact polygon clipping (a 12-, 20- and 28-gon around c_N^* of areas 17/189, 61/693 and 437/4977, and the four points of Q_N), independently of the induction; a referee's own program confirmed Proposition 9 with equality for $k \leq 4$, and found no exception among 10^7 sampled points for $k = 1, 2$. Exact certificates (branch and bound over the integer values of a basis of the relations, with a rational Farkas vector at every leaf), each accepted by two checkers written separately (`check_w.py` and `check_w_indep.py`), show that the sets of Theorem 4 have no such character: in case (1), with the N that the proof needs, $d = 23, 47, 71, 95$ with $N = 25$ and

$d = 119, 143, \dots, 311$ ($d \equiv 23 \pmod{24}$, squarefree) with $N = 125$, an independent check of Section 4; in case (2), already with a smaller N than the proof needs, $d = 11, 35, 59$ with $N = 25$ and $d = 83, 131, 155$ with $N = 125$. A floating-point solver confirms case (2) with the N that the proof needs for $d = 11, 35, 59, 83, 131, 179$. Finally, for a finite set Δ of unit vectors, Proposition 9 and Lemma 10 reduce the question whether $G_N(\Delta \cup \{1\})$ admits such a character for all large N to the existence of a character with values in $\{\frac{1}{3}, \frac{1}{2}, \frac{2}{3}\}$, a computation in $(\mathbb{Z}/6)^4$; it confirms Theorem 4(2) for all 191 squarefree $d \equiv 11 \pmod{24}$ below 5000, and shows that for $d = 11$ no two vectors $(\mu + i\nu\sqrt{11})/\tau$ with $\nu \leq 7$, $\tau \leq 60$ suffice. Independently of the proof, a computer search over denominators found a certificate of the same kind, for other sets of unit vectors, for each of the 155 squarefree $d \equiv 11 \pmod{12}$ below 2000 and for 593 of the 611 between 2000 and 10 000; both checkers accept all of them, except two of more than 50 MB, which were checked once and not kept (`data/quadratic_planes/winding/scan` in [13]).

Formal proof. Theorem 1 and Corollary 3 are also proved in Lean 4 with Mathlib (`lean/FourColours.lean` and `lean/PadicFour.lean` in [13], the values $\chi(\mathbb{Q}_2^2) = 2$ and $\chi(\mathbb{Q}_3^2) = 3$ in `lean/PadicPlanes.lean`, and Theorem W for an arbitrary abelian group and a finite set S in `lean/TheoremWInf.lean`, by averages over boxes and a limit along an ultrafilter); the files depend only on Lean’s three standard axioms. The formal proof uses a cruder form of Proposition 9, which suffices for Theorem 1 but not for the bound $d < 21 \cdot 5^{k-1}$: every $c \in S_N$ is $N(\frac{1+i}{2} + t) + x$ with $t \in \mathbb{Z}[i]$ and $|x|^2 < \frac{1}{18}$, or $N(\frac{\alpha+\beta i}{3} + t)$ with $\alpha, \beta \in \{1, 2\}$. This follows by the induction of Proposition 9 started at $k = 0$, where $S_1 = ([\frac{1}{3}, \frac{2}{3}] + \mathbb{Z})^2$ and the corners of the square are the points of type q ; in the step, $\frac{1}{18} = (\frac{7}{30})^2 + (\frac{1}{30})^2$ replaces the numbers of Lemma 7, and the points of type q are handled by a congruence modulo 15. The formal analogue of Lemma 10 assumes $N > 2 \sum_j |m_j|$, for values written as $\frac{N}{6}E + x$ with E an integer and $|x| \leq \frac{1}{4}$.

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